(b) An analysis of the properties and complexity (time, space, sample size) of any algorithm. [Yes, the time complexity and data sizes are given in Appendix E.2]

(c) (Optional) Anonymized source code, with specification of all dependencies, including external libraries. [Yes, our code is anonymized.]

2. For any theoretical claim, check if you include:

(a) Statements of the full set of assumptions of all theoretical results. [Yes, all our theoretical statements contain the full set of assumptions, see Appendix D.]

(b) Complete proofs of all theoretical results. [Yes, all proofs are given in Appendix D.]

(c) Clear explanations of any assumptions. [Yes, we explain our assumptions throughout the paper.]

3. For all figures and tables that present empirical results, check if you include:

(a) The code, data, and instructions needed to reproduce the main experimental results (either in the supplemental material or as a URL). [Yes, the full experimental details are provided in Appendix E.]

(b) All the training details (e.g., data splits, hyperparameters, how they were chosen). [Yes, all training details are provided in Appendix E.]

(c) A clear definition of the specific measure or statistics and error bars (e.g., with respect to the random seed after running experiments multiple times). [Yes, measures and statistics are provided in Section 6 and in Appendix E.]

(d) A description of the computing infrastructure used. (e.g., type of GPUs, internal cluster, or cloud provider). [Yes, see AppendixE.4.]

4. If you are using existing assets (e.g., code, data, models) or curating/releasing new assets, check if you include:

(a) Citations of the creator If your work uses existing assets. [Yes, we cited existing assets throughout the paper.]

(b) The license information of the assets, if applicable. [Yes, all assets used are publicly available.]

(c) New assets either in the supplemental material or as a URL, if applicable. [Yes, our code is provided in the supplemental material.]

(d) Information about consent from data providers/curators. [Not Applicable, all assets used are publicly available.]

(e) Discussion of sensible content if applicable, e.g., personally identifiable information or offensive content. [Yes, see Section 7.]

5. If you used crowdsourcing or conducted research with human subjects, check if you include:

(a) The full text of instructions given to participants and screenshots. [Not Applicable, we did not use crowdsourcing.]

(b) Descriptions of potential participant risks, with links to Institutional Review Board (IRB) approvals if applicable. [Not Applicable, we did not use crowdsourcing.]

(c) The estimated hourly wage paid to participants and the total amount spent on participant compensation. [Not Applicable, we did not use crowdsourcing.]

A Additional Related Work

The methods introduced in our paper draw on ideas from survival analysis, conformal prediction, and black-box uncertainty quantification. The `Optimized` method (Section 5.2) in particular is also related to: conformal prediction foundations, uncertainty quantification for LLMs, active learning and statistical inference, and optimal experimental design.

Conformal Prediction. The theoretical foundations of conformal prediction were established by Kolmogorov (1983, 1968); Papadopoulos et al. (2002); Vovk et al. (2005). This framework has since been extended to accommodate covariate shift by applying weights to the distribution of conformity scores (Tibshirani et al., 2019). Similar weighting strategies have also been employed to integrate active learning into conformal prediction (Prinster et al., 2024). Building further on these weighted approaches, conformal prediction methods have recently been adapted for survival analysis tasks (Candès et al., 2023; Gui et al., 2024a; Davidov et al., 2025).

Uncertainty Quantification for LLMs. Several recent works adapt conformal ideas to off-the-shelf LLMs. For instance, Wang et al. (2024) adapts conformal prediction to open-ended generation by integrating a self–consistency–based uncertainty measure, achieving strict correctness coverage across seven popular LLMs and diverse domains. The work of Wang et al. (2025) further extends this by explicitly inserting the ground truth into candidate outputs and using logit-based nonconformity scores to guarantee coverage over a wider range of error rates, though its reliance on logits raises calibration concerns when only API-level access is available. Additionally, *Conformal Language Modeling* has been developed by Quach et al. (2024) to calibrate both stopping and rejection rules to guarantee coverage in open-domain QA and summarization. Lin et al. (2024) distinguish uncertainty from confidence using semantic dispersion metrics, enabling selective generation of low-confidence outputs. To refine validity under practical constraints, Cherian et al. (2024) extended conditional conformal methods to incorporate utility-based guarantees and differentiable filtering. Mohri and Hashimoto (2024) introduced "conformal factuality," constructing entailment-based uncertainty sets that deliver high-probability correctness. The approach proposed in (Gui et al., 2024b) provides a guarantee for aligned responses in QA and radiology applications.

Active Learning and Statistical Inference. Active learning methods aim to select the most informative samples for labeling, improving model accuracy under tight budgets (Settles, 2010). This was formalized in (Zrnic and Candès, 2024) for statistical inference tasks, where one prioritizes labeling points with high model uncertainty while maintaining valid confidence intervals. Our `Optimized` method relates to this idea as it derives sample probabilities that directly minimize the variance of the augmented inverse-propensity-weighting estimator.

Optimal Experimental Design. Optimal experimental design allocates observations to minimize estimator variance under a given model. In (Hahn et al., 2011), the authors develop a two-stage adaptive design that chooses propensity scores to minimize the asymptotic variance bound in treatment-effect estimation, and the approach presented in (List et al., 2011) presents simple rules of thumb for efficient designs under resource constraints. Analogously, our `Optimized` method treats the sampling rule $\pi(x)$ as a design variable and solves for the allocation that minimizes the miscoverage estimation variance.

Classical Survival Analysis. The survival analysis problem was first formalized in biostatistics and reliability engineering. Kaplan and Meier (1958) introduce a nonparametric estimator of the survival function for right-censored data, laying the groundwork for all subsequent methods. The proportional hazards model was later developed, formalizing a semiparametric regression framework that relates covariates to an event's hazard rate without specifying its baseline form. As an alternative formulation, accelerated failure time (AFT) (Lin et al., 1998) models assume a parametric form for the log-survival time, allowing direct modeling of time-to-event via common distributions such as Weibull or log-normal. More recently, machine-learning approaches have been developed for survival prediction. Random survival forests leverage ensemble tree methods to nonparametrically estimate cumulative hazard functions (Ishwaran et al., 2008). DeepSurv employs a neural-network approximation of the Cox partial likelihood to capture complex covariate interactions (Katzman et al., 2018), and Deep Cox Mixtures generalizes this by modeling mixtures of proportional hazards to flexibly adapt to heterogeneous subpopulations (Nagpal et al., 2021).

B Calibration Algorithms

B.1 Calibration With a Known Censoring Mechanism

We present the general calibration with known censoring mechanism in Algorithm 1.

Algorithm 1 Calibration With a Known Censoring Mechanism

Require: Calibration data $\{X_i, C_i, \tilde{T}_i\}_{i\in\mathcal{I}_2}$, censoring weights $\{w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)\}_{i\in\mathcal{I}_2,\tau\in\mathcal{T}}$, quantile estimates $\{\hat{f}_\tau(\cdot)\}_{\tau\in\mathcal{T}}$, target miscoverage rate α, prior quantile τ_{prior}.

for $\tau \in \mathcal{T} \cap [0, \tau_{\text{prior}}]$ **do**

  $\hat{\alpha}(\tau) \leftarrow \frac{1}{|\mathcal{I}_2|} \sum_{i\in\mathcal{I}_2} w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)\, \mathbb{I}\{\tilde{T}_i < \hat{f}_\tau(X_i) \leq C_i\}$ // miscoverage est.

end for

$\hat{\tau} \leftarrow \sup\Big\{\tau \in \mathcal{T} \cap [0, \tau_{\text{prior}}] : \sup_{\tau'\in\mathcal{T}, \tau'\leq\tau} \hat{\alpha}(\tau') \leq \alpha\Big\}$ // calibrated quantile level

return Lower predictive bound (LPB) for a test point $X_{\text{test}} = x$, given by $\hat{L}(x) = \hat{f}_{\hat{\tau}}(x)$

B.2 Calibration With a Known Censoring Mechanism: `Naive` Mode

We describe the `Naive` calibration approach in Algorithm 4. First, we provide two auxiliary algorithms to sample outcomes from a generative model.

Algorithm 2 Generate and Audit Generative Model Responses

Require: Calibration data $\{X_i\}_{i\in\mathcal{I}_2}$, Generative model $\mathcal{G}(\cdot)$, audit function `Audit`$(\cdot)$, Censoring times $\{C_i\}_{i\in\mathcal{I}_2}$.

for each $i \in \mathcal{I}$ **do**

  Initialize $j \leftarrow 0$

  repeat

    $j \leftarrow j + 1$

    Generate and evaluate the output's safety: $Y_i^j \leftarrow$ `Audit`$(X_i, \mathcal{G}(X_i))$

  until $Y_i^j = 1$ **or** $C_i = j$

  Set $\tilde{T}_i \leftarrow j$

end forreturn $\{\tilde{T}_i\}_{i\in\mathcal{I}_2}$

Algorithm 3 Generate Generative Model Outcomes

Require: Calibration data $\{X_i\}_{i\in\mathcal{I}_2}$, generative model $\mathcal{G}(\cdot)$, max per-prompt budget $\{\hat{f}_{\tau_{\text{prior}}}(X_i)\}_{i\in\mathcal{I}}$, per-prompt evaluation probability $\{\pi_i\}_{i\in\mathcal{I}}$, audit function `Audit`$(\cdot)$.

for each $i \in \mathcal{I}$ **do**

  Draw $V_i \sim$ Bernoulli(π_i)

  Set $C_i \leftarrow V_i \cdot \hat{f}_{\tau_{\text{prior}}}(X_i)$

end for

Obtain $\{\tilde{T}_i\}_{\{i\in\mathcal{I}_2\}}$ from Algorithm 2 applied with $\{X_i\}_{i\in\mathcal{I}_2}$, $\mathcal{G}$, $\{C_i\}_{i\in\mathcal{I}_2}$. **return** $\{(\tilde{T}_i, C_i)\}_{i\in\mathcal{I}_2}$.

B.3 Calibration With a Known Censoring Mechanism: `Basic` Mode

The calibration procedure with `Basic` budget allocation is outlined in Algorithm 5.

B.4 Calibration With a Known Censoring Mechanism: `Trimmed` Mode

We describe the calibration procedure with trimmed budget allocation in Algorithm 6.

Algorithm 4 Calibration With a Known Censoring Mechanism: `Naive` mode

Require: Calibration data $\{X_i\}_{i\in\mathcal{I}_2}$, generative model $\mathcal{G}(\cdot)$, audit function `Audit`$(\cdot)$, pre-trained quantile regression model $\{\hat{q}_\tau(\cdot)\}_{\tau\in\mathcal{T}}$, target miscoverage rate α, prior quantile τ_{prior}, total budget B.

Sample $C_i \sim \text{Geom}(n/B)$ for all $i \in \mathcal{I}_2$

Obtain $\{\tilde{T}_i\}_{\{i\in\mathcal{I}_2\}}$ from Algorithm 2 applied with $\{X_i\}_{i\in\mathcal{I}_2}$, $\mathcal{G}$, $\{C_i\}_{i\in\mathcal{I}_2}$.

$w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i) \leftarrow \frac{1}{\mathbb{P}[\hat{q}_\tau(X_i)\le C_i|X_i]},\ i \in \mathcal{I}_2$. // compute the weights for the geometric C_i.

$\{(\tilde{T}_i, C_i)\}_{i\in\mathcal{I}_2} \leftarrow$ Algorithm 3 applied to $\{X_i\}_{i\in\mathcal{I}_2}$ with $\{\pi_i\}_{i\in\mathcal{I}_2}$ and $\{\hat{f}_{\tau_{\text{prior}}}(X_i)\}_{i\in\mathcal{I}_2}$

$w(\{X_j\}_{j\in\mathcal{I}_2}, i) \leftarrow \frac{1}{\pi_i},\ i \in \mathcal{I}_2$

Obtain $\hat{L}(x)$ from Algorithm 1, applied with $\{X_i, C_i, \tilde{T}_i\}_{i\in\mathcal{I}_2}$, $\{w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)\}_{i\in\mathcal{I}_2}$, $\{\hat{q}_\tau(\cdot)\}_{\tau\in\mathcal{T}}$, α, and τ_{prior}.

return $\hat{L}(x)$, the lower predictive bound (LPB) for a test point $X_{\text{test}} = x$.

Algorithm 5 Calibration With a Known Censoring Mechanism: `Basic` mode

Require: Calibration data $\{X_i\}_{i\in\mathcal{I}_2}$, generative model $\mathcal{G}(\cdot)$, audit function `Audit`$(\cdot)$, pre-trained quantile regression model $\{\hat{q}_\tau(\cdot)\}_{\tau\in\mathcal{T}}$, target miscoverage rate α, prior quantile τ_{prior}, total budget B.

$\{\hat{f}_\tau(X_i)\}_{\tau\in\mathcal{T},i\in\mathcal{I}_2} \leftarrow \{\hat{q}_\tau(X_i)\}_{\tau\in\mathcal{T},i\in\mathcal{I}_2}$

$\{\pi_i\} \leftarrow \{\min(\frac{B}{|\mathcal{I}_2|\ \hat{f}_{\tau_{\text{prior}}}(X_i)}, 1)\}$ // per-prompt evaluation probability (Section 6.1)

$\{(\tilde{T}_i, C_i)\}_{i\in\mathcal{I}_2} \leftarrow$ Algorithm 3 applied to $\{X_i\}_{i\in\mathcal{I}_2}$ with $\{\pi_i\}_{i\in\mathcal{I}_2}$ and $\{\hat{f}_{\tau_{\text{prior}}}(X_i)\}_{i\in\mathcal{I}_2}$

$w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i) \leftarrow \frac{1}{\pi_i},\ i \in \mathcal{I}_2$

Obtain $\hat{L}(x)$ from Algorithm 1, applied with $\{X_i, C_i, \tilde{T}_i\}_{i\in\mathcal{I}_2}$, $\{w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)\}_{i\in\mathcal{I}_2}$, $\{\hat{q}_\tau(\cdot)\}_{\tau\in\mathcal{T}}$, α, and τ_{prior}.

return $\hat{L}(x)$, the lower predictive bound (LPB) for a test point $X_{\text{test}} = x$.

Algorithm 6 Calibration With a Known Censoring Mechanism: `Trimmed` Mode

Require: Calibration data $\{X_i\}_{i\in\mathcal{I}_2}$, generative model $\mathcal{G}(\cdot)$, audit function `Audit`$(\cdot)$, pre-trained quantile regression model $\{\hat{q}_\tau(\cdot)\}_{\tau\in\mathcal{T}}$, target miscoverage rate α, prior quantile τ_{prior}, total budget B.

$\{\hat{f}_\tau(X_i)\}_{\tau\in\mathcal{T},i\in\mathcal{I}_2} \leftarrow \{\hat{q}_\tau(X_i)\}_{\tau\in\mathcal{T},i\in\mathcal{I}_2}$

$\{\pi_i\} \leftarrow \{\min(\frac{B}{|\mathcal{I}_2|\ \hat{f}_{\tau_{\text{prior}}}(X_i)}, 1)\}$ // per-prompt evaluation probability (Section 6.1)

$\{(\tilde{T}_i, C_i)\}_{i\in\mathcal{I}_2} \leftarrow$ Algorithm 3 applied to $\{X_i\}_{i\in\mathcal{I}_2}$ with $\{\pi_i\}_{i\in\mathcal{I}_2}$ and $\{\hat{f}_{\tau_{\text{prior}}}(X_i)\}_{i\in\mathcal{I}_2}$

$w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i) \leftarrow \frac{1}{\pi_i},\ i \in \mathcal{I}_2$

Obtain $\hat{L}(x)$ from Algorithm 1, applied with $\{X_i, C_i, \tilde{T}_i\}_{i\in\mathcal{I}_2}$, $\{w_\tau(\{X_j\}_{j\in\mathcal{I}_2}, i)\}_{i\in\mathcal{I}_2}$, $\{\hat{q}_\tau(\cdot)\}_{\tau\in\mathcal{T}}$, α, and τ_{prior}.

return $\hat{L}(x)$, the lower predictive bound (LPB) for a test point $X_{\text{test}} = x$.